Recall

Work from memory, then check with your tutor.

R1. Find 140% of 30 points.

Answer: _____

R2. A rectangle has area $\frac{3}{4}$ square meter and width $\frac{3}{4}$ meter. Find its length.

Answer: _____

R3. Find 2.8 ÷ 0.07.

Answer: _____

R4. A snail moves 3 centimeters each minute. Find the distance after 0, 4, and 7 minutes.

Answer: _____

S. Return here near the end of the lesson

I can check a solution by __.

Equations and related quantities

Follow the slide labels. P = problem; F = your turn; D = discussion.

P1. Solve x + 7 = 19. Show the same change on both sides.

Answer: _____

P2. Four identical bags hold 28 marbles. Write and solve an equation for marbles per bag, b.

Answer: _____

P3. Solve 1.2x = 5.4. Check with multiplication.

Answer: _____

Equations and related quantities

Follow the slide labels. P = problem; F = your turn; D = discussion.

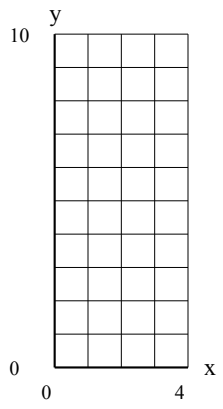
P4. Solve $(\frac{2}{3})x = 10$. Draw three equal parts of x first.

Answer: _____

P5. A game costs $18. Mina has $6.50 saved. Write and solve an equation for the extra amount, a, she needs.

Answer: _____

P6. Notebooks cost $2.50 each. Use c = 2.5n for n = 0, 1, 2, 4. Plot (n, c). Name the input and output.



x: notebooks n; y: cost c

Equations and related quantities

Follow the slide labels. P = problem; F = your turn; D = discussion.

F. Complete: 5x = 35. Divide both sides by __. Then x = __.

Answer: _____

D. Omar solves 6x = 42 by writing x = 42 − 6 = 36. Explain and repair his work.

Answer: _____

Mixed review

M1. 24 shells are 30% of a collection. Find the whole collection.

Answer: _____

M2. Find $2 \div \frac{2}{3}$.

Answer: _____

M3. Find 3.45 + 5.078.

Answer: _____

Exit ticket

Work on your own. Show a model or reason when the question asks for one.

E. Solve $(\frac{3}{5})x = 12$. Show equal parts or an equation step. Check your value.

Answer: _____

Expressions and equations

The big idea

Expressions name quantities. Equivalent forms describe the same quantity. Equations can be solved by keeping both sides equal.

Words to know

variable: A letter that can stand for a number. Example: In 5n, n can be 2 or 7.

expression: Numbers, variables, and operations without an equals sign. Example: 5n + 3

exponent: The number of equal factors to multiply. Example: 4^2 means 4 × 4.

equivalent: Having the same value for every allowed input. Example: 2(x + 5) and 2x + 10

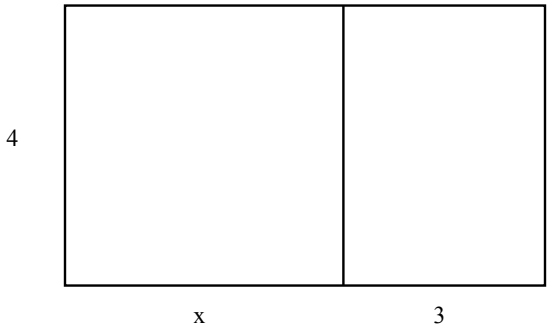
equation: A statement that two quantities are equal. Example: x + 7 = 19

solution: A value that makes an equation true. Example: 12 solves x + 7 = 19.

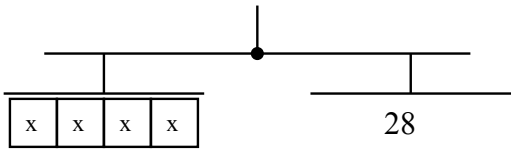
independent variable: The input you choose. Example: Number of notebooks, n.

dependent variable: The output determined by the input. Example: Total cost, c = 2.5n.

How it works



Four rows each contain x + 3. An expression names a quantity. Equivalent expressions name the same quantity for every allowed value.



4 equal bags 28 marbles

Four equal unknown groups balance 28. A solution makes the equation true. The same operation on both sides preserves equality.

Worked examples

1. Find 3(2 + 4).

The parentheses name one group: 2 + 4 = 6.

Three groups give 3 × 6 = 18.

Split each group instead: 3 × 2 + 3 × 4 = 6 + 12 = 18.

Answer: 18

2. Solve x + 2.75 = 8.

Both sides must stay equal.

Subtract 2.75 from each side: x = 5.25.

Check: 5.25 + 2.75 = 8.

Answer: x = 5.25

Common mistakes

Writes $4^2 = 8$. Why: Reads the exponent as multiplication by two. Fix: Write two factors of 4, then compare.

Writes 4(x + 3) = 4x + 3. Why: Loses the four groups of 3. Fix: Shade the numeric part in all four rows.

Checks one value and claims equivalence. Why: A single match can be accidental. Fix: Ask for an area argument that works for any x.

Try it

1. Expand $2(x + 7)$.

2. Expand $3(x + 6)$.

3. Solve $x + 9 = 23$.

4. Solve $x + 14 = 31$.

5. Solve $x + 11 = 18$.

Answers

1. $2x + 14$ 2. $3x + 18$ 3. $x = 14$ 4. $x = 17$ 5. $x = 7$

Where this connects

Ratios and fraction division help solve equations. These same equations will describe proportional relationships.